Area Optimization After Technology Mapping

Under a fixed delay bound

Shared logic, local rewriting, and reusable timing profiles

Why optimize after mapping?

Logic representation

- AND-inverter graph (AIG)
- Node count and logic depth

Mapped implementation

- Standard-cell area and pin delays
- Shared cells and explicit inversions

$$\text{Fewer AIG nodes} \not\Rightarrow \text{smaller standard-cell area}$$

A local rewrite saves area only when its old cells become removable.

Related work: [1] post-mapping optimization, [2] technology mapping and graph optimization.

Problem formulation

$$\min_{N} A(N) \quad \text{subject to} \quad D(N) \leq D_0,\ N \equiv N_0$$

$$A(N) = \sum_{v \in V_{\text{cell}}(N)} a(v), \qquad D_0 = D(N_0)$$

- N_0: initial mapped netlist. Keep D_0 fixed across rounds
- Minimize area first. Use delay to break area ties
- Allow local slack consumption within the global bound

Area: μm^2. Delay: ps, using library pin delays. Physical load and slew are outside this model.

Design questions

Structure	Which equivalent implementations expose useful sharing?
Cost	What area and delay will a replacement have in the live netlist?
Search	How much candidate construction is affordable?

Method: 16 min **Experiments: 9 min** **Discussion: 2 min**

Two optimization scopes

Across roots

Compose AO21 chain blocks.

Share factors and segments.

Evaluate one joint transaction.

Within one root

Enumerate structural cuts.

Generate and price covers.

Use reverse topological order.

Common guard: rebuilt area and the original delay bound

A round starts from a checkpoint. Whole-window synthesis and unmap/remap exploration are disabled.

AO21 chain composition

An AO21 step represents $F_i(z) = (p_i \wedge z) \vee q_i$.

$$
\begin{aligned}
F_1(F_2(z)) &= p_1 \wedge (p_2 \wedge z \vee q_2) \vee q_1 \\
&= \underbrace{(p_1 \wedge p_2)}_{P} \wedge z \;\vee\; \underbrace{(q_1 \vee p_1 \wedge q_2)}_{Q}
\end{aligned}
$$

$$
(P, Q) \circ (p, q) = (P \wedge p,\; Q \vee P \wedge q)
$$

Exact Boolean identity, including dependent coefficient signals. Current configuration: two-step blocks, minimum chain length six.

Sharing changes the unit of evaluation

$$y_r = P \wedge z_r \vee Q_r$$

$$y_s = P \wedge z_s \vee Q_s$$

$P = p_1 \wedge p_2$ is built once for both roots.

- Product memo: sorted, unique factors
- Segment memo: ordered coefficient pairs
- Rebuild the joint proposal to count all surviving cells

Accept the transaction only when $A_{\text{new}} < A_{\text{old}}$ and $D \leq D_0$

Local rewriting on structural cuts

$$\mathcal{C} = (x, L), \qquad |L| \leq 6$$

Boundary

- x: root to replace
- L: signals cutting every input-to-root path
- Current arrivals belong to leaf signals

Function

- Truth table over L
- Conservative care set
- A replacement agrees on care minterms

Care-window simulation identifies unreachable minterms. Cofactor candidates and e-graph search use the full function. The window itself is not synthesized.

Complementary candidate sources

Generator	Candidate structure
L1: single cell	Function matching with phases and pin assignments
L2: database	Pre-mapped multi-cell directed acyclic graphs (DAGs)
L3: synthesis	Factoring, disjoint-support decomposition, and cofactors
L4: e-graph	Equivalent expressions with library-cell covers

L1–L3 compete first. L4 is a fallback on the MFFC boundary cut.

MFFC: maximum fanout-free cone. L4 triggers only after no L1–L3 proposal passes the acceptance filters on that cut.

Cofactor decompositions

For a pivot ℓ, define $f_b = f|_{\ell=b}$ and $\Delta_\ell = f_0 \oplus f_1$.

$f = (\ell \wedge f_1) \vee (\neg\ell \wedge f_0)$	Shannon
$f = f_0 \oplus (\ell \wedge \Delta_\ell)$	Positive Davio
$f = f_1 \oplus (\neg\ell \wedge \Delta_\ell)$	Negative Davio

Equivalent functions can produce different standard-cell covers

Each form is factored, mapped, and verified independently on the complete cut truth table.

Pivot selection under a search budget

Let $s(g)$ denote the number of support variables in g.

Family	Score to minimize	Retained pivots
Shannon	$s(f_0) + s(f_1)$	2
Positive Davio	$s(f_0) + s(\Delta_\ell)$	2
Negative Davio	$s(f_1) + s(\Delta_\ell)$	2

- Resolve ties by leaf order, then process retained pivots in original order
- Check timing after mapping

Support size is a heuristic. The cap can omit useful covers. The all-pivot ablation measures that tradeoff.

Bounded equality saturation

An e-class records equivalent expressions [3]

1. Expand the cut into an AIG with fixed leaf order
2. Apply Boolean rewrites and library matching
3. Initialize one area-oriented realization per e-class
4. Widen the retained frontier within the remaining budget

Iterations	e-nodes	Covers/class	Combined extraction budget
2	500	up to 8	20,000 combinations / 4,000 visits

Both extraction passes share one budget. Finite rules and frontiers give an incomplete candidate pool.

Timing profiles for reusable covers

$$T(H;\tau) = \max_{i \in U_H} \{\tau(\ell_i) + p_i(H)\}$$

$p_i(H)$: input-to-root path delay. U_H: leaves used by H.

Cover	p_0	p_1	T at $(0,0)$	T at $(20,0)$
H_1	10	1	10	30
H_2	1	10	10	21

Store $\left(\widehat{A}(H), p_0, \ldots, p_{|L|-1}\right)$, then select using live arrivals.

Illustrative equal-area covers, delays in ps. Scalar zero-arrival delay loses this distinction. Bounded pruning uses the complete delay vector.

Area cost in the live netlist

$$u = \neg v \qquad \Longrightarrow \qquad \neg u = v$$

$$\widehat{A}_{\mathrm{ctx}}(H) = \underbrace{\sum_{v \in V_H^{\mathrm{new}}} a(v)}_{\text{new reachable cells}} + \underbrace{\sum_{v \in J_H \cap M} a(v)}_{\text{old cells retained by reuse}}$$

$$\Delta\widehat{A}(H) = \widehat{A}_{\mathrm{ctx}}(H) - \sum_{v \in M} a(v)$$

Resolve external phases before computing both area and timing.

M: removable old cells. J_H: reused old cells. Existing shared cells outside M add no area. Reused signals must not depend on the replacement root.

Local feasibility and global acceptance

Local selection

Evaluate live leaf arrivals.

Require $T(H) \leq R(x)$.

Rank by $\Delta\widehat{A}$, then arrival.

Global guard

Check function and cycles.

Invalidate affected caches.

Rebuild and validate (A, D).

$$R(o) = D_0, \qquad R(u) = \min_{(u,v,i)} \{R(v) - d_{v,i} - b_{v,i} d_{\text{inv}}\}$$

R: required arrival time. $d_{v,i}$: pin delay. $b_{v,i}$: input-inversion indicator. A round rolls back if it violates D_0 or worsens the area-first objective.

Reusing work and restricting search

Reuse computed results

- Inverse-phase queries
- Truth-table simulation
- Mapped DAGs for identical AIG states

Recompute live cost and timing. Invalidate records when their context changes.

Restrict candidate coverage

- Two pivots per family
- Bounded saturation
- Limited extraction frontiers

Search limits can omit useful replacements.

Report solution quality, completion, and runtime separately

Statistical deferrals are revisited. Only strict area bounds skip live pricing among generated covers.

Experimental protocol

Item	Setting
Designs	16 fixed circuits with ASAP7 library models
mfs3 comparison	Same initial netlist, same 40-entry library (≤ 6 inputs)
Other experiments	Full 47-entry library, fixed original AIG inputs
Post-mapping budget	1000 s for both tools
Disabled operations	Whole-window synthesis and unmap/remap exploration

The six-input restriction comes from the tested mfs3 implementation. Preprocessing, initial mapping, and combinational equivalence checking (CEC) are outside the post-mapping budget.

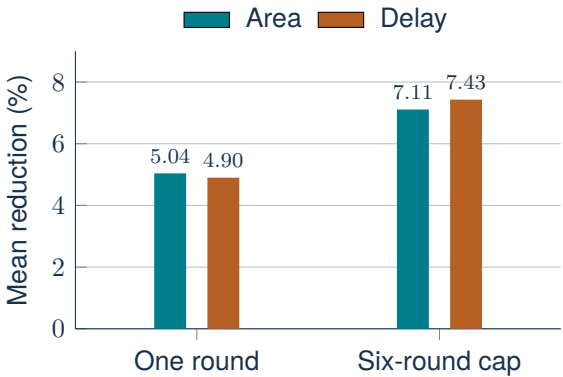
Evidence from three experiments

Experiment	Starting netlist	Evidence sought
Strong AIG optimization	N_e, round caps 1 and 6	Remaining area opportunity
mfs3 (ABC [4])	Same N_i, one round	Same-input tool comparison
Mechanism ablations	Same N_e, six-round cap	Quality and runtime tradeoffs

N_i: direct mapping.
N_e: mapping after high-effort AIG preprocessing.

ASPDAC'25 [1] has no available source implementation in this study. Its published costs are reconstructed separately, with different mapped inputs.

Area gains after strong AIG optimization



16 of 16
Smaller area, with $D \leq D_0$.

3.59%
Mean area reduction excluding arbiter.

Arithmetic mean of per-circuit percentage reductions. Caps 1 and 6 are independent runs. Cold and warm modes yield identical PPA.

arbiter: a structural case study

	Initial	Optimized
Area (μm^2)	557.72	223.64
Delay (ps)	999.87	261.39

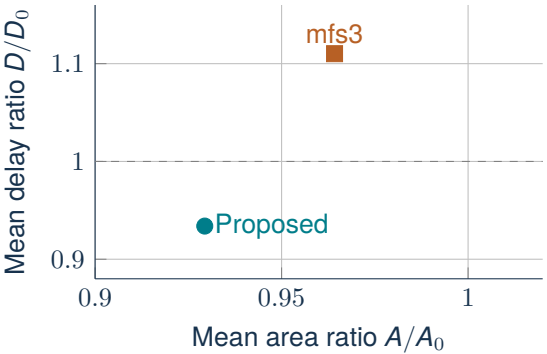
Same original input

Chain composition exposes repeated factors across roots.

$$\underbrace{557.72 - 223.64}_{\text{area reduction}} = 334.08\ \mu\text{m}^2$$

Removing chain composition or cross-root sharing gives a summed area penalty of $+334.08$ in the ablations. Mechanism effects can interact.

Same-input comparison with mfs3



Completion

ABC: 13/16
Proposed: 16/16

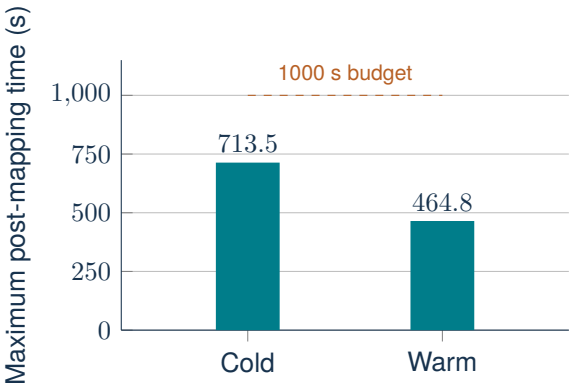
Delay violations

ABC: 7 completed cases
Proposed: 0

Raw area: 7 wins, 6 losses versus ABC on completed pairs.

Points are means over the 13 ABC-completed pairs, using each tool's native imported baseline. mfs3 -a does not enforce D_0. Dashed line: unit mean delay ratio.

Completion and cache-dependent runtime



128/128 completed

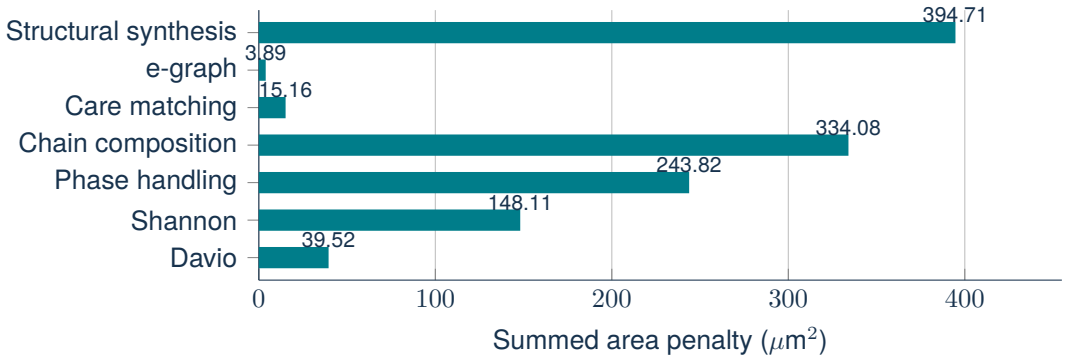
Full configuration only.

Paired mean (s)	Time
Proposed, cold	75.0
Proposed, warm	27.8
ABC	26.4

Cold: empty profile pool. Warm: corresponding cold pool.

Deadline: 1000 s. 64 runs per mode across four full-configuration settings. Native caches restart. OS caches remain. Shared-host, single-run timing. Controls have 7 TOs.

Area penalties from removing mechanisms



$\Sigma\Delta A = \sum_d (A_{d,\text{disabled}} - A_{d,\text{full}})$. Same area penalties in cold and warm modes. All shown controls complete. Effects are not additive. Full runtime data are in the appendix.

Reuse and search coverage: different tradeoffs

Control	$\Sigma\Delta A$	Time/full	TOs
No query reuse	+69.76 / +68.63	1.67 / 2.15	2 / 2
No table reuse	+0.00 / +0.00	1.02 / 1.02	0 / 0
All pivots	-36.93 / -39.99	1.28 / 1.55	1 / 1

- Query reuse: same PPA on 14 completed pairs
- Table reuse: same PPA on all 16 circuits
- All pivots: extra area savings, with mem_ctrl timing out

Pairs show cold / warm. Area differences in μm^2. Query control also removes the level shortcut. Table control retains scalar cut memoization. Time includes TO budgets, so ratios are not equal-work speedups.

Scope of the evidence

Supported by these experiments	Remaining limitations
Area gains at fixed initial delay	Library timing, without physical sign-off
Budget completion on the tested suite	Finite rules and frontiers can miss covers
Same-input mfs3 comparison	Different internal delay objectives
Cold/warm runtime observations	Shared host and no repeated-run statistics

ASPDAC'25 literature reconstruction: 14/16 smaller final areas, but different mapped inputs. This does not isolate a post-mapping algorithm advantage.

Conclusions

1. Mapped-cell structure retains area opportunities after AIG optimization
2. Cross-root sharing and local covers address different scopes
3. Reusable profiles require live timing and sharing evaluation

16/16 main circuits: smaller area, no delay regression

Open question
How should circuit structure determine the search budget?

Appendix: main results, 1/2

Circuit	A_0	A	D_0	D	t_c/t_w (s)
aes_core	1029.98	996.04	410.72	407.66	420.6/156.4
arbiter	557.72	223.64	999.87	261.39	19.5/19.7
div	1184.50	1182.38	53105.06	53024.38	210.2/170.1
ethernet	2541.41	2531.03	543.88	543.56	466.8/330.6
iwls05_i2c	47.66	45.11	303.61	290.02	18.8/4.9
mem_ctrl	1895.40	1818.16	1519.75	1461.58	713.5/463.9
RISC	2927.19	2807.40	1453.77	1392.91	517.4/267.8
sasc	27.28	26.96	191.20	191.20	3.0/1.4

Full library; six-round cap. Area in μm^2, delay in ps; same initial netlists; identical cold/warm PPA.

Appendix: main results, 2/2

Circuit	A_0	A	D_0	D	t_c/t_w (s)
simple_spi	36.96	35.89	308.22	293.36	8.0/3.1
spi	149.43	146.68	500.98	450.90	57.0/21.1
sqrt	1157.38	1071.21	88015.07	80961.45	221.8/206.7
systemcaes	483.75	447.57	709.03	682.83	112.8/58.1
systemcdes	131.06	125.89	488.85	487.52	75.8/25.0
tv80	340.61	330.87	846.12	833.94	159.6/63.8
usb_funct	646.23	622.62	668.08	664.39	207.9/105.6
usb_phy	22.54	21.39	163.29	158.68	9.3/1.9

Full library; six-round cap. Area in μm^2, delay in ps; same initial netlists; identical cold/warm PPA.

Appendix: same-input ABC comparison, 1/2

	ABC mfs3			Proposed method		
Circuit	A/A_0	D/D_0	t (s)	A/A_0	D/D_0	t_c/t_w (s)
aes_core	0.9731	1.1333†	32.5	0.9825	0.9999	186.0/59.7
arbiter	1.0000	1.0000	96.3	0.6282	0.5655	6.6/6.5
div	TO	TO	1000	0.9606	0.9954	407.1/342.4
ethernet	0.9935	1.1019†	52.6	0.9863	0.9812	150.8/69.8
iwls05_i2c	0.9601	0.9843	0.9	0.9259	0.9901	14.8/2.4
mem_ctrl	TO	TO	1000	0.9591	0.9896	376.2/174.6
RISC	0.9715	1.6875†	91.9	0.9446	0.9977	341.8/128.2
sasc	0.9816	1.0000	0.1	0.9886	1.0000	1.6/0.8

Same $\leq$ 6-input library and initial netlist. Native baseline normalization; †: delay violation; TO: 1000 s. CEC excluded.

Appendix: same-input ABC comparison, 2/2

	ABC mfs3			Proposed method		
Circuit	A/A_0	D/D_0	t (s)	A/A_0	D/D_0	t_c/t_w (s)
simple_spi	0.9730	1.0000	0.1	0.9766	0.9022	6.6/1.4
spi	0.9786	1.0510†	7.6	0.9550	0.9912	24.4/8.1
sqrt	TO	TO	1000	0.8857	0.9996	90.1/80.9
systemcaes	0.9704	1.0904†	17.6	0.9753	0.8132	35.0/17.3
systemcdes	0.9251	1.2116†	1.8	0.9382	0.9947	23.3/13.0
tv80	0.9301	1.1757†	38.4	0.9491	0.9805	85.9/26.6
usb_funct	0.9745	0.9968	3.5	0.9482	0.9392	93.8/26.5
usb_phy	0.9035	1.0000	0.1	0.8832	0.9852	4.3/1.1

Same ≤ 6-input library and initial netlist. Native baseline normalization; †: delay violation; TO: 1000 s. CEC excluded.

Appendix: complete ablations, 1/2

Control	$\Sigma\Delta A$ C/W	Time/full C/W	TOs C/W
full	+0.00/+0.00	1.00/1.00	0/0
noSYN	+394.71/+394.71	0.73/0.59	0/0
noEGG	+3.89/+3.89	0.51/0.86	0/0
noDB	+1.62/+1.62	0.82/0.71	0/0
noCARE	+15.16/+15.16	0.95/0.91	0/0
noCHAIN	+334.08/+334.08	1.00/0.99	0/0
noSHARE	+334.08/+334.08	1.00/1.00	0/0
noNARROW	+0.92/+0.92	1.15/1.24	0/0
noPHASE	+243.82/+243.82	0.88/0.74	0/0
noSHANNON	+148.11/+148.11	0.97/0.95	0/0

C/W: cold/warm. Area difference in μm^2, positive is worse. Verified timeout snapshots retain incomplete status.

Appendix: complete ablations, 2/2

Control	$\Sigma\Delta A$ C/W	Time/full C/W	TOs C/W
noDAVIO	+39.52/+39.52	0.97/0.92	0/0
noREUSE	+231.82/+231.82	1.01/0.92	0/0
noPRICE	+163.59/+163.59	0.82/0.69	0/0
cache64K	+0.28/+0.00	1.18/1.37	1/0
noAIGREUSE	+0.00/+0.00	1.00/0.99	0/0
noLIBSHARE	+0.00/+0.00	1.03/1.03	0/0
noQUERY	+69.76/+68.63	1.67/2.15	2/2
noTRUTH	+0.00/+0.00	1.02/1.02	0/0
allPIVOTS	-36.93/-39.99	1.28/1.55	1/1

C/W: cold/warm. Area difference in μm^2, positive is worse. Verified timeout snapshots retain incomplete status.

Appendix: parameters and validation

- Cuts: $K = 6$, 12 cuts per root; e-graph threshold $\theta = 0$
- Care: 8-input reconvergent window; full care after 60 s; SAT off
- Database: three levels, area cap 0.50; native cache: 262144 entries
- Profiles: 2 iterations, 500 nodes, 8 covers, 20000 combinations, 4000 visits
- 2 pivots per family; SOP/DSD: ≥ 4 leaves; Shannon/Davio: ≥ 3
- Periodic legal snapshots; independent CEC and serialized PPA audit for TOs
- Shared host: up to 12 method jobs and 6 ABC jobs, one observation per case

Appendix: references

[1] A. Costamagna et al. *Area-Oriented Optimization After Standard-Cell Mapping*. ASPDAC, 2025. DOI: 10.1145/3658617.3697722.

[2] A. Tempia Calvino et al. *A Versatile Mapping Approach for Technology Mapping and Graph Optimization*. ASPDAC, 2022.

[3] M. Willsey et al. *egg: Fast and Extensible Equality Saturation*. PACMPL/POPL, 2021.

[4] R. Brayton and A. Mishchenko. *ABC: An Academic Industrial-Strength Verification Tool*. CAV, 2010.

Experimental data: fixed 16-circuit campaign. Main and ablation tables use the same admitted output records. Source hashes accompany the LaTeX data snapshot.